Standard Operating Procedure

Installing WSL 2 and Ubuntu on Windows 11

A Linux Practice Environment for a Windows Home PC

Field	Value
Operating System	Windows 11 Home Single Language
Architecture	x64-based PC
Computer Name	PMSQLDBA
RAM	24 GB
Processor	Intel Core i5-1135G7
Target Linux Distribution	Ubuntu 26.04 LTS (via WSL 2)
Cygwin Source	https://cygwin.com/install.html
Starting State	No Cygwin, no WSL, no Linux distribution installed

Document owner: IT / DBA Training Desk

Target machine: PMSQLDBA

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Reference sources: Microsoft Learn, Install WSL and Basic Commands for WSL documentation.

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1. Purpose

This SOP gives a beginner a precise, repeatable procedure for setting up a working Linux environment on a Windows 11 home PC using Windows Subsystem for Linux 2 (WSL 2) and Ubuntu. Follow each step in order. Do not skip the restart step.

2. Scope and Current System State

This procedure applies to a Windows 11 Home PC with no Linux tooling installed. Cygwin is not installed, WSL is not installed, and no Linux distribution is present. The table below lists the exact system this SOP targets and the starting point for the setup.

Field	Value
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Target Linux Distribution	Ubuntu 26.04 LTS (via WSL 2)
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Starting State	No Cygwin, no WSL, no Linux distribution installed

Status check: Nothing is installed yet. Cygwin installs first from the official source, then WSL and Ubuntu install through PowerShell, then Windows restarts to activate the new components.

3. Advantage of This Setup Over Installing a Full Linux OS

A beginner does not need to install a separate Linux operating system, partition a disk, or run a dual-boot setup to practice Linux and Ubuntu commands. WSL 2 delivers a full Linux kernel and Ubuntu userspace directly inside Windows 11. The advantages of this approach are listed below.

- No disk partitioning. The Windows drive stays untouched, so there is no risk of breaking the existing Windows installation.
- No dual boot. Windows and Ubuntu run at the same time, so switching back and forth does not require a restart.
- Fast setup. Ubuntu installs in minutes through a single command, compared to hours for a full OS install.

- Low resource cost. WSL 2 uses a lightweight virtual machine, so a 24 GB RAM laptop runs Windows and Ubuntu together without strain.
- Full package manager access. apt, sudo, and standard Ubuntu repositories work exactly as they do on a physical Linux server.
- Real Linux kernel. Commands such as lsblk, ps, and uname report genuine Linux kernel data, not an emulation layer.
- Safe to reset. Ubuntu on WSL uninstalls and reinstalls cleanly with a single command if something breaks during practice.
- Shared file access. Windows files stay reachable from Ubuntu through /mnt/c, so files move between both systems without extra tools.
- Ideal for DBA and Linux administration training. lsblk, df, ps, systemctl-related tooling, network commands, and package management work the same way a junior Linux administrator uses them on a live server.

In short, a Windows Home PC user gains a genuine Linux administration and DBA practice environment without buying separate hardware, installing a hypervisor, or reformatting a drive.

4. Environment Map: Cygwin, WSL 2, and PowerShell

Three separate environments exist on this PC. Confusing them is the most common beginner mistake. The table below states the purpose of each one.

Tool	Purpose	Use It For
Cygwin	POSIX-style shell layered on Windows	Bash scripting, Git, SSH, grep, awk, sed
WSL 2 + Ubuntu	Real Linux kernel and userspace inside Windows	lsblk, apt, sudo, systemd tools, Linux administration practice
PowerShell	Native Windows administration shell	Get-Disk, Get-Partition, Get-Process, WSL management commands

Cygwin provides a bash-style shell without a Linux kernel underneath it. WSL 2 provides an actual Linux kernel and, once installed, the Ubuntu userspace on top of it. PowerShell manages Windows itself and also manages WSL from the Windows side.

5. Prerequisites

- Windows 11 with administrator rights on the account performing the setup
- Active internet connection for the Cygwin download and the Ubuntu package download
- At least 5 GB of free disk space for the Ubuntu filesystem, plus space for the Cygwin install
- Access to https://cygwin.com/install.html to download the Cygwin installer

6. Step-by-Step Installation Procedure

Step 1: Download Cygwin from the Official Source

Cygwin is not installed on this PC yet. Download it directly from the official site before touching WSL.

1. Open a web browser and go to https://cygwin.com/install.html.
2. Download setup-x86_64.exe, the 64-bit Cygwin installer for this PC.
3. Save the file to a folder such as Downloads.

> **Source:** https://cygwin.com/install.html is the only official download location for the Cygwin installer. Avoid third-party mirrors.

Step 2: Install Cygwin

Run the downloaded installer and complete the setup wizard.

1. Double-click setup-x86_64.exe.
2. Choose Install from Internet on the setup type screen.
3. Accept the default root install directory, for example C:\cygwin64.
4. Choose a local package download directory, for example Downloads.
5. Pick a nearby download mirror from the site list.
6. On the package selection screen, keep the default package set. To add Git, OpenSSH, and common utilities now, search each package name and select the newest version.
7. Click Next and let the installer finish downloading and installing packages.
8. Click Finish to close the wizard.

A Cygwin Terminal shortcut now appears on the desktop and in the Start menu. Open it and confirm a prompt such as the one below appears.

```
DELL@PMSQLDBA ~$
```

Cygwin is now installed. WSL and Ubuntu install next.

Step 3: Open PowerShell as Administrator

WSL installs through PowerShell, not through Cygwin.

1. Click Start.
2. Type PowerShell.
3. Right-click Windows PowerShell.
4. Select Run as administrator.

Use PowerShell for every command in Step 4 and Step 5. Do not run these commands inside Cygwin.

Step 4: Install WSL and Ubuntu

WSL is not installed on this PC yet. The command below installs the WSL components, enables the Virtual Machine Platform Windows feature, sets WSL 2 as the default version, and installs Ubuntu in a single step.

```
wsl --install -d Ubuntu
```

Expect output similar to the lines below while the components and the Ubuntu package download.

```
Installing: Windows Subsystem for Linux
Installing: Virtual Machine Platform
Installing: Ubuntu
The operation completed successfully.
The requested operation is successful. Changes will not be effective until
the system is rebooted.
```

If the download stalls or fails, run the command below instead, which pulls the package from an alternate source.

```
wsl --install --web-download -d Ubuntu
```

Step 5: Restart Windows 11

The install output states that changes take effect only after a restart. Save open work, then restart through Start, Power, Restart. Do not proceed to Step 6 before the restart finishes.

Step 6: Verify WSL Installed Successfully

After Windows restarts, open PowerShell as Administrator again and confirm the install.

```
wsl --status
wsl --list --verbose
```

wsl --status confirms WSL is active. wsl --list --verbose lists Ubuntu with VERSION 2. Both results confirm the install finished correctly.

Step 7: Create the Linux Username and Password

Ubuntu opens automatically after the restart to finish setup. If it does not open on its own, launch it from the Start menu or run wsl -d Ubuntu from PowerShell. Ubuntu then prompts for a new account.

1. At the prompt Enter new UNIX username, type a lowercase username, for example dell.
2. At the prompt New password, type a password. The screen shows nothing while typing. This is normal Linux behavior, not an error.
3. Retype the password to confirm it.

Beginner tip: The Linux username does not need to match the Windows username. They stay independent of each other.

Step 8: Verify the Ubuntu Session

Inside the new Ubuntu window, run each command below and compare the result to the expected output.

```
whoami
id
pwd
```

whoami returns the Linux username. pwd returns a path such as /home/dell, confirming a genuine Linux filesystem.

Step 9: Confirm the Distribution and Kernel

```
cat /etc/os-release
uname -a
```

cat /etc/os-release identifies the distribution as Ubuntu. uname -a reports a kernel string containing microsoft-standard-WSL2, confirming Ubuntu runs on the real WSL 2 Linux kernel.

Step 10: Test Core Linux Commands

```
lsblk
df -h
free -h
ps aux
ip addr
ss -tuln
```

Each command now runs as a genuine Linux utility. lsblk lists block devices as seen by the WSL virtual machine, not the raw physical Windows disk layout. For the physical Windows disk layout, run Get-Disk and Get-Partition from PowerShell instead.

Step 11: Test sudo and the Root Account

```
sudo -i
whoami
exit
```

sudo -i switches to the root account after the Linux password is entered. whoami then returns root. exit returns to the normal user account.

Step 12: Update Ubuntu

```
sudo apt update
sudo apt upgrade -y
```

This refreshes the package index and applies available updates through the standard Ubuntu package manager.

Step 13: Install Core Administration Tools

```
sudo apt install -y \
    vim nano curl wget git unzip zip tree \
    htop net-tools dnsutils lsof rsync \
    openssh-client sysstat
```

This installs the standard toolset for Linux administration and DBA practice, including network, process, and performance utilities.

Step 14: Set WSL 2 and Ubuntu as Defaults

Run these commands from PowerShell, not Ubuntu.

```
wsl --set-default-version 2
wsl --set-default Ubuntu
```

This sets WSL 2 as the default version for future distributions and Ubuntu as the default distribution for the wsl command.

Step 15: Launch Ubuntu Going Forward

```
wsl
wsl -d Ubuntu
wsl -d Ubuntu -u root
```

The first command opens the default distribution. The second targets Ubuntu directly. The third opens Ubuntu as the root account for recovery work.

Step 16: Access Windows Files from Ubuntu

```
cd /mnt/c
ls
cd /mnt/c/Users/DELL
```

Windows drives sit under /mnt inside Ubuntu. This path reaches the Windows C: drive and the Windows user folder directly from Linux.

Step 17: Verify Network Connectivity

```
ip addr
ip route
ping -c 4 8.8.8.8
ping -c 4 google.com
```

A successful reply from both ping commands confirms Ubuntu has working internet access and DNS resolution.

7. Verification Checklist

Confirm every item below before marking the setup complete.

- Cygwin downloaded from https://cygwin.com/install.html and installed
- PowerShell opened with administrator rights
- wsl --install -d Ubuntu completed without error
- Windows 11 restarted after the WSL install
- wsl --status confirms WSL is active
- Ubuntu appears in wsl --list --verbose
- Ubuntu shows VERSION 2 in the same output
- Linux username and password are set
- cat /etc/os-release reports Ubuntu
- uname -a reports a microsoft-standard-WSL2 kernel
- lsblk runs inside Ubuntu without error
- sudo -i switches to the root user
- sudo apt update and sudo apt upgrade -y complete without error
- Windows files are reachable at /mnt/c
- Network connectivity is confirmed with ping and DNS lookup

8. Environment Recognition Quick Reference

A beginner mixes up terminals often. Use the prompt shape below to identify the active environment at a glance.

Cygwin prompt

```
DELL@PMSQLDBA ~$
```

Ubuntu (WSL 2) prompt

```
dell@PMSQLDBA:~$
```

PowerShell prompt

```
PS C:\Users\DELL>
```

A Linux command such as lsblk only runs correctly inside the Ubuntu prompt. Running it inside Cygwin returns command not found, because Cygwin has no Linux kernel underneath it.

9. Troubleshooting

lsblk: command not found

This means the command ran inside Cygwin or PowerShell instead of Ubuntu. Open Ubuntu with wsl -d Ubuntu, then retry.

Sudo is disabled on this machine message

This message comes from Windows itself, not Ubuntu. It appears when sudo runs inside Cygwin or PowerShell. Inside Ubuntu, sudo works normally after the Linux password is entered.

wsl --install -d Ubuntu hangs or fails to download

Run the web download variant instead.

```
wsl --install --web-download -d Ubuntu
```

lsblk output does not match the physical disk

WSL 2 runs Ubuntu inside a lightweight virtual machine, so lsblk reports virtual disks, not the physical Windows NVMe or SSD layout. Check the physical layout from PowerShell instead.

```
Get-Disk
Get-Partition
```

10. Completion Criteria

Setup is complete once every item in Section 7 is checked and Ubuntu responds correctly to the commands in Steps 8 through 17.

Reference:

Microsoft Learn, Install WSL (learn.microsoft.com/windows/wsl/install) and Basic commands for WSL (learn.microsoft.com/windows/wsl/basic-commands).